# Venkata Lakshmi Parimala Pasupuleti

Principal GenAI & Agentic Systems Engineer · Python · LangGraph · LLM Orchestration · Microservices · On-Prem + Cloud LLMs · Financial Data at Scale

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**Hiring Team**
Bank of America — Global Markets GenAI Technology
Re: Principal Engineer — GenAI Team

## COVER LETTER

Global Markets GenAI Technology is building a platform that combines LLM reasoning with structured and unstructured financial datasets to power insights across Sales & Trading and quantitative strategy groups. That intersection — **agentic AI systems running on financial data at scale** — is exactly where my work sits. **AgentProbe** is a 3-agent LangGraph runtime with tool calling, multi-provider LLM routing across cloud (Claude, OpenAI, Groq) and **on-premises (Ollama)**, a 5-metric eval engine with CI/CD regression gates on every commit, and a React observability dashboard tracking retrieval drift and latency automatically. **Leasel** achieves 93.6% precision on 1,240 unstructured legal documents with a deterministic rules engine that validates every LLM output before it touches the database. Both are shipped and benchmarked — not whiteboard designs.

On production scale: at Entrata I shipped the company's **first production AI pipeline across 1.6M financial records** (Python, PyTorch, OpenAI API), led a **microservices architecture redesign** (50% scalability improvement, 30% deployment reduction), optimized **200+ SQL queries** (30% performance gain) across all client environments, and maintained 99.5% uptime across 100K+ monthly active users over 6+ years. Building GenAI systems is one skill. Building GenAI systems that stay up, scale, and handle real financial data without corruption is another — that's what 9+ years of production engineering buys.

One capability I'd specifically highlight for a Global Markets context: AgentProbe's multi-provider routing includes **Ollama for on-premises LLM inference**, designed precisely for environments where sensitive financial data cannot leave the network. The eval engine can compare outputs across cloud and on-prem providers within the same pipeline — a meaningful capability when your data includes proprietary trading signals. I'm open to relocating to New York and look forward to discussing the role.

Sincerely,

**Venkata Lakshmi Parimala Pasupuleti**